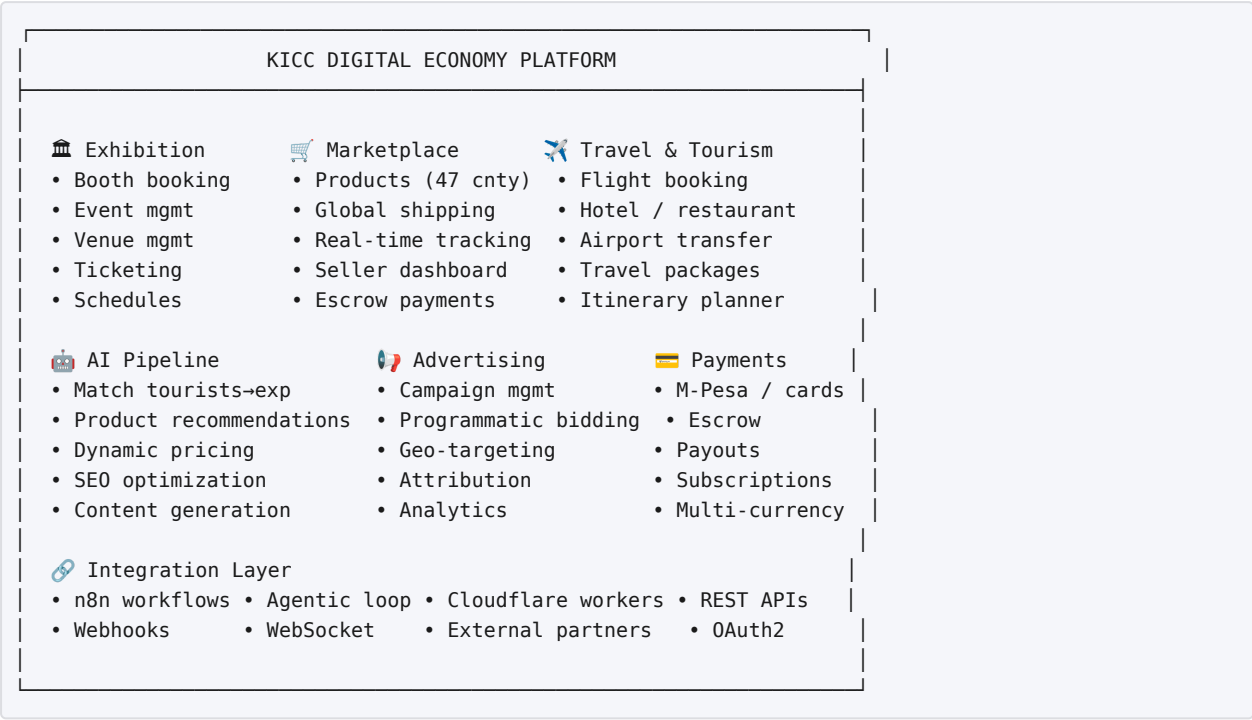


KICC Digital Economy Platform — Architecture Blueprint

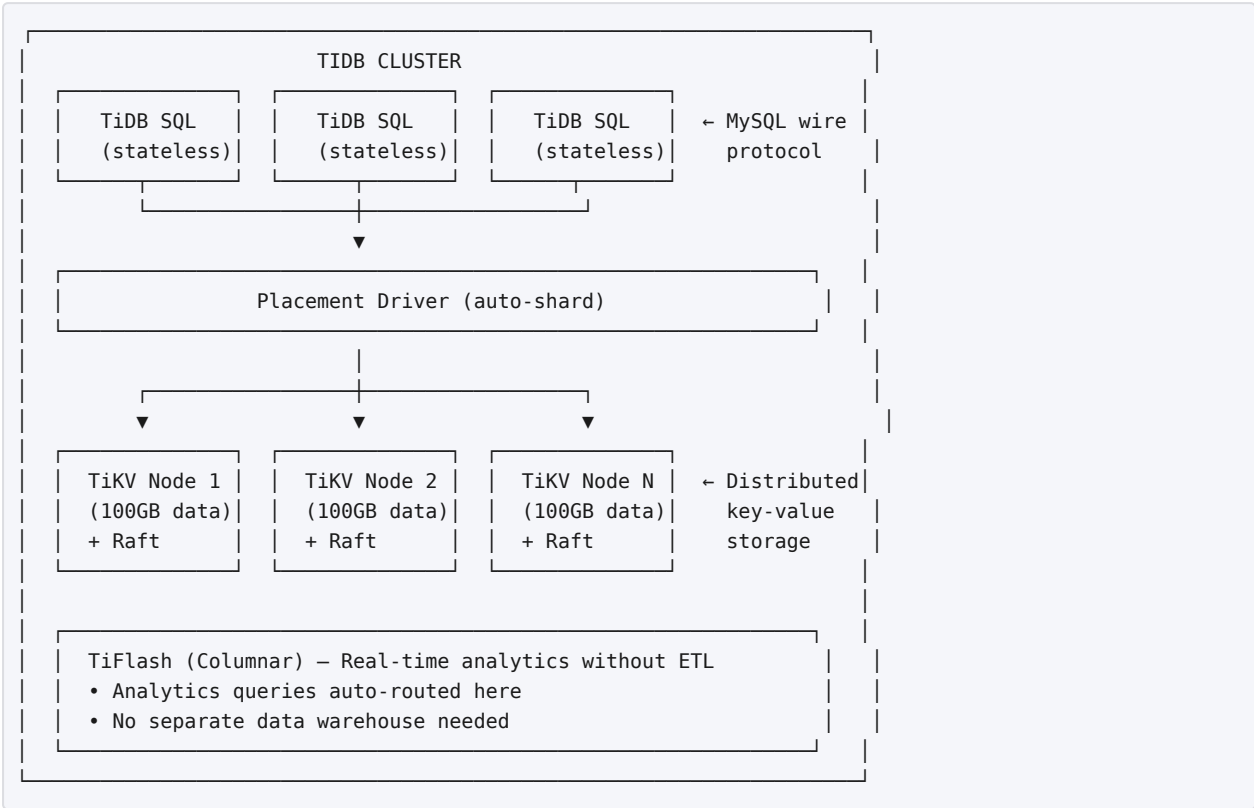
Platform Vision

A unified **Kenyan Digital Economy Super-App** connecting:



Architecture Layers

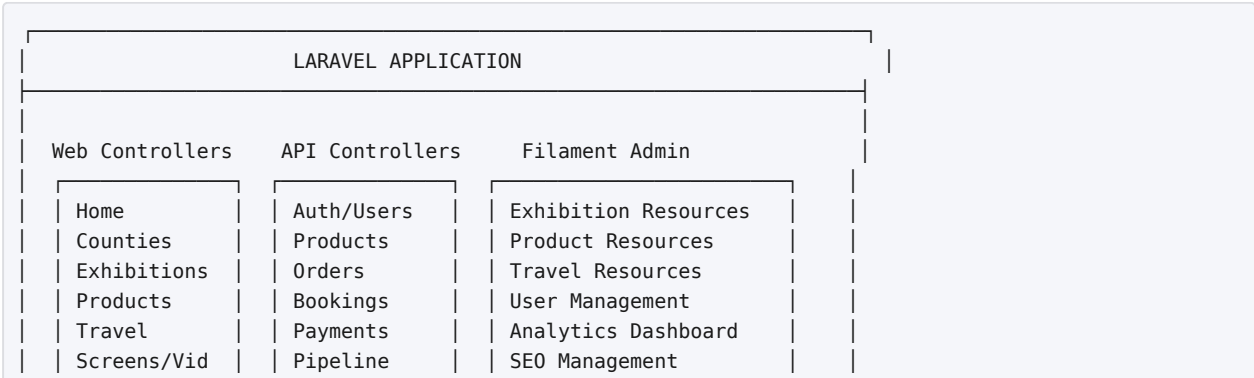
Layer 1: Database (TiDB — MySQL Compatible Distributed)

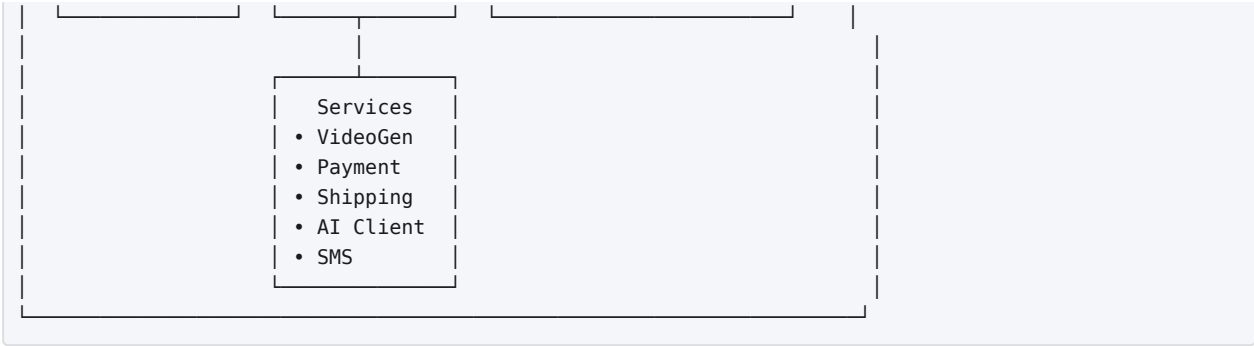


Why TiDB over Aurora MySQL:

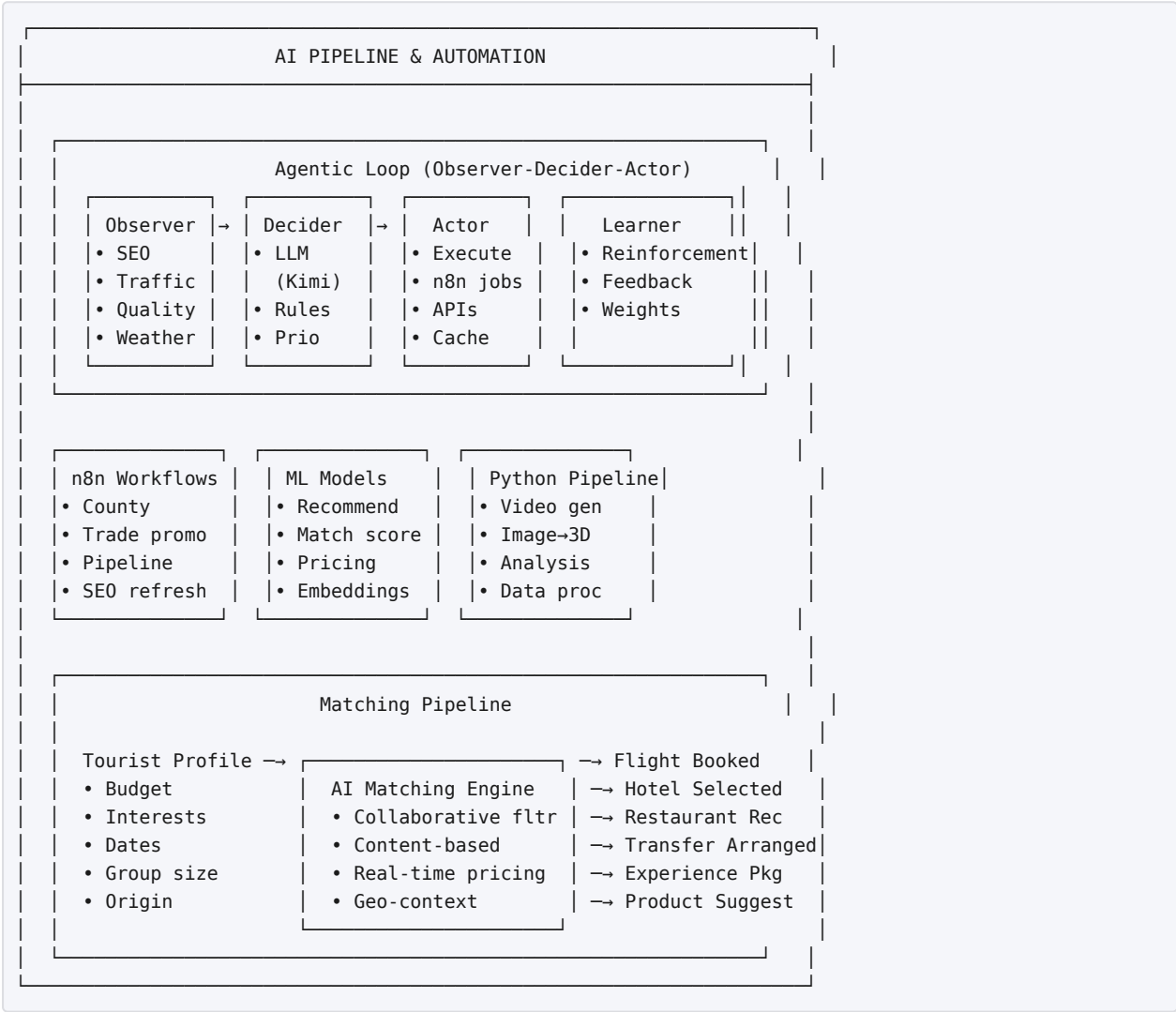
Capability	Aurora MySQL	TiDB
Max storage	256 TiB (single node write bottleneck)	Petabyte+ (auto-shards across nodes)
Write scaling	Single primary node	Multi-node writes — linear scaling
Multi-region	Read replicas only	Active-active across regions
HTAP	Separate Redshift/Snowflake needed	Built-in columnar (TiFlash) — analytics on live data
MySQL compat	✅ Full	✅ Wire-protocol compatible
Zero-downtime scaling	Vertical (resize instance)	Horizontal (add nodes, no downtime)
Cost at petabyte scale	Very expensive (large instances)	Commodity hardware — scales linearly

Layer 2: Application (Laravel)

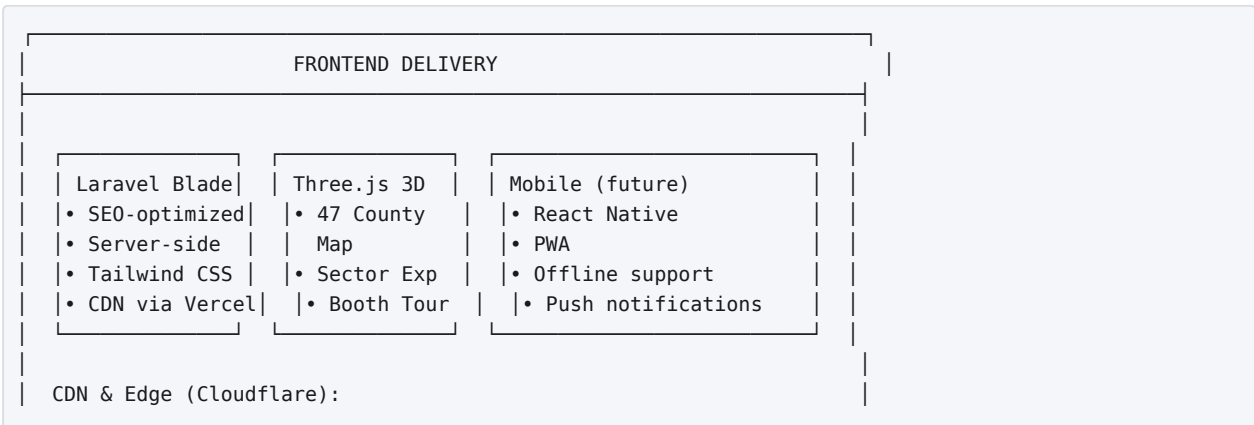




Layer 3: AI Pipeline & Automation



Layer 4: Frontend Delivery



- Static assets • 3D HTML • Videos • Images • API caching
- DDoS protection • SSL • Workers for geo-routing

Domain Model — 20 Business Domains (~200 tables)

Core Platform (56 existing tables — kept as-is)

E-Commerce Marketplace (new)

Table	Purpose
products	All Kenyan products across 47 counties
product_categories	Category tree (agri, crafts, textiles, etc.)
product_variants	Size, color, weight variants
product_images	Gallery photos
product_reviews	Buyer ratings
suppliers	County-based suppliers
supplier_products	Supplier→product mapping
inventory	Stock per warehouse/variant
warehouses	Fulfillment centers
shopping_carts	Active carts (per user)
cart_items	Items in cart
orders	Customer orders
order_items	Line items
order_status_history	State machine log
shipments	Fulfillment records
shipment_tracking	Courier tracking events
return_requests	RMA process
wishlists	Saved items
coupon_codes	Discount campaigns
price_history	Pricing analytics

Travel & Tourism (new)

Table	Purpose
airlines	Partner airlines
airports	All Kenyan airports
flights	Flight schedules/routes
flight_inventory	Seat availability
flight_bookings	Booking records
flight_passengers	Passenger manifests
hotels	Partner hotels
hotel_rooms	Room types/inventory
hotel_bookings	Booking records
hotel_room_bookings	Room→booking mapping
restaurants	Partner restaurants
restaurant_tables	Table layouts
restaurant_menu_items	Menu catalog
restaurant_bookings	Table reservations
airport_transfers	Transfer services
transfer_bookings	Pickup/dropoff records
travel_packages	Bundled experiences
travel_itineraries	Day-by-day plans
attractions	Tourist sites
attraction_tickets	Entry tickets

AI Pipeline (new)

Table	Purpose
pipeline_jobs	Job definitions
pipeline_runs	Execution history
pipeline_logs	Step-by-step logs
ml_models	Model registry
model_versions	Version tracking
embeddings	Vector embeddings (product, user, content)
recommendations	Generated recommendations
recommendation_logs	Impression/click/conv logging
user_segments	ML-generated segments
match_scores	User→item match scores
a_b_tests	Experiment tracking
feature_flags	Rollout management

Advertising (new)

Table	Purpose
ad_campaigns	Campaign definitions
ad_groups	Ad group hierarchy
ad_creatives	Image/video/text ads
ad_targeting	Audience targeting rules
ad_placements	Slot definitions
ad_impressions	Impression log (high volume)
ad_clicks	Click log
ad_conversions	Conversion tracking
ad_budgets	Budget pacing
ad_publisher_payouts	Revenue share

SEO & Content (new)

Table	Purpose
seo_metadata	Per-page SEO metadata
content_pages	CMS pages
content_blocks	Reusable content
redirects	URL redirect rules
sitemap_urls	Sitemap tracking
search_queries	Internal search log
keyword_tracking	Search ranking history

Logistics & Shipping (new)

Table	Purpose
courier_partners	Shipping carriers
shipping_zones	Regional zones
shipping_rates	Rate cards
shipping_labels	Generated labels
pickup_requests	Supplier pickup
delivery_attempts	Failed delivery log

Payments (expanded)

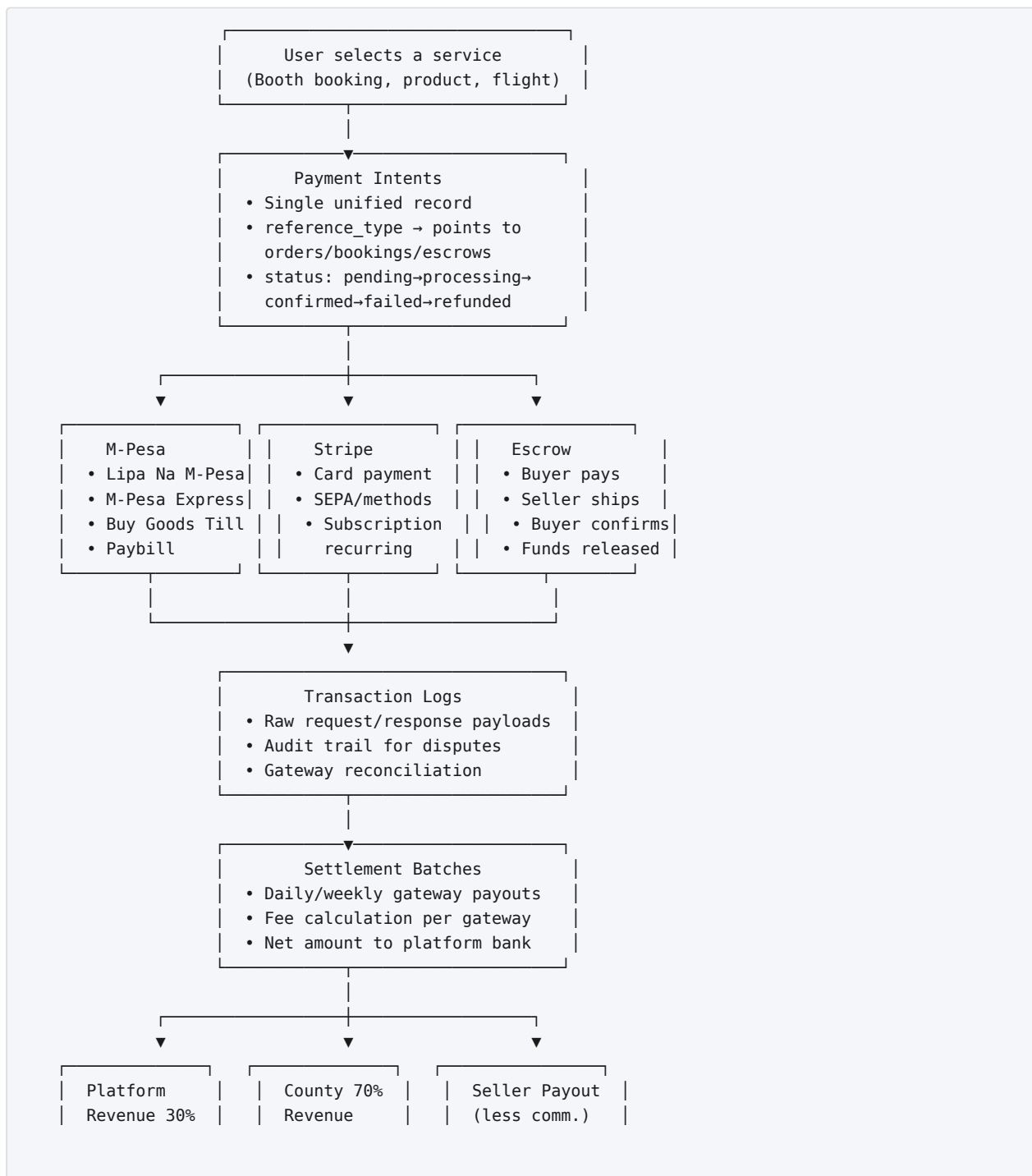
Table	Purpose
payment_gateways	Gateway configs (M-Pesa, Stripe, etc.)
payment_intents	Unified payment record across all domains
transaction_logs	Raw gateway request/response payloads
refunds	Full/partial refund processing
payment_disputes	Buyer/seller dispute resolution
seller_payouts	Bulk payout to suppliers (weekly/monthly)
settlement_batches	Gateway settlement reconciliation
settlement_transactions	Individual txns in a settlement batch
tax_rates	VAT, withholding tax configs per product/county
currency_rates	KES→USD→EUR→GBP exchange rates
advertiser_transactions	Ad account top-ups and spend

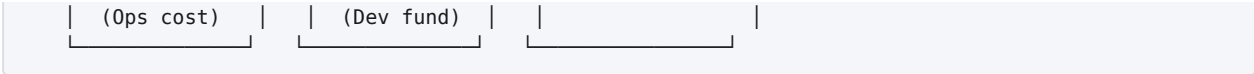
Subscriptions (expanded)

Table	Purpose
subscription_features	Feature catalog (max_booths, analytics, etc.)
subscription_plan_features	Plan→feature value mapping
usage_logs	Metered API/feature usage per subscriber
billing_cycles	Monthly/yearly billing periods
invoices	Generated invoice records
invoice_items	Individual line items per invoice

Payments & Subscriptions — Deep Dive

Payment Flow Architecture





3-Tier Subscription Model

Tier	Target	Price (KES)	Key Features	County Benefit
Free	All users	0	1 booth listing, basic profile	County gets free visibility
County Premium	County govt	50,000/mo	20 booths, analytics, SEO boost, priority support	County promotes all local businesses
Exhibitor Pro	Businesses	5,000/mo	5 booths, analytics, livestream, multi-event	County earns 30% commission
Enterprise	Corporates	50,000/mo	Unlimited booths, API access, white-label, dedicated support	County earns 25% commission

County Revenue Model

Each county has its own financial configuration in TiDB:

```
county_financial_config:
├─ revenue_share_pct: 70      # 70% to county, 30% to platform
├─ mpesa_paybill: "123456"    # County-specific M-Pesa till
├─ settlement_period: "monthly" # Monthly/weekly payouts
├─ county_wallet_balance: 0    # Running balance in KES
└─ subscription_discount: 0.20 # 20% discount for county residents
```

How money flows to counties:

```
Product sold (1,000 KES)
├─ Platform fee (10%): 100 KES
├─ Payment gateway (3%): 30 KES
├─ County revenue share: 609 KES (70% of remaining 870)
└─ Seller nets: 261 KES (30% of remaining 870)
```

Revenue sources per county:

Source	County Share	Platform Share
Booth bookings	70%	30%
Marketplace sales	70%	30%
Travel commissions	50%	50%
Ad revenue (local)	80%	20%
Subscription fees	30%	70%
Escrow fees	50%	50%

Reconciliation & Settlement

```
Daily cutoff (23:59 EAT)
├─ Aggregate all intents by county
├─ Deduct gateway fees + platform commission
├─ Generate settlement_batch per county
├─ Auto-transfer to county M-Pesa paybill
│   (or accumulate for monthly lump sum)
└─ Log in county_wallet_transactions table
```

Subscription + Usage Metering

```

User signs up for "County Premium" (50,000 KES/mo)
|
|— billing_cycles created (monthly periods)
|— invoice generated
|— payment_intent via M-Pesa/Stripe
|
After payment:
|— subscription_plan_features grants:
|   • max_booths = 20
|   • has_analytics = true
|   • has_livestream = true
|
usage_logs tracks:
|— Booth slots used (counted against max)
|— Analytics queries this month
|— API calls
---

## Database Recommendation: **TiDB**

**TiDB is the optimal choice** for this platform because:

1. **MySQL wire-protocol compatible** – Your existing Laravel + 56 tables work with **zero code changes**. Same Eloquent queries, same migrations, same everything.

2. **Auto-sharding (no sharding logic)** – TiDB automatically distributes data across nodes. When you grow from 10 GB to 10 TB to 1 PB, you just add nodes. No manual partitioning, no Vitess complexity.

3. **Petabyte-scale HTAP** – TiFlash columnar replicas let you run real-time analytics directly on production data. No ETL to Snowflake/Redshift. Analytics queries auto-route to TiFlash.

4. **Multi-region active-active** – Deploy in US-East, EU-West, Asia-Pacific simultaneously. Users get local read/write with automatic conflict resolution.

5. **Linear write scaling** – Unlike Aurora's single-primary bottleneck, TiDB writes scale linearly with nodes. 100M transactions/day? Add nodes.

6. **Zero-downtime operations** – Add indexes, change schema, scale up/down without locking. Critical for a 24/7 marketplace.

7. **Cost at scale** – Commodity hardware. 10 nodes of 8 CPU/32GB RAM can handle 1 TB+ for ~$500/month on bare metal, much less than Aurora at that scale.

### Migration Path

```

SQLite (local dev) → TiDB Serverless (free tier, 5GB) → TiDB Dedicated (prod)

1. **Local:** SQLite (already configured) – for development
2. **Staging:** TiDB Serverless (free up to 5GB, no server to manage)
3. **Production:** TiDB Dedicated cluster (auto-scale, multi-region)

Migration is a single `.env`` change:

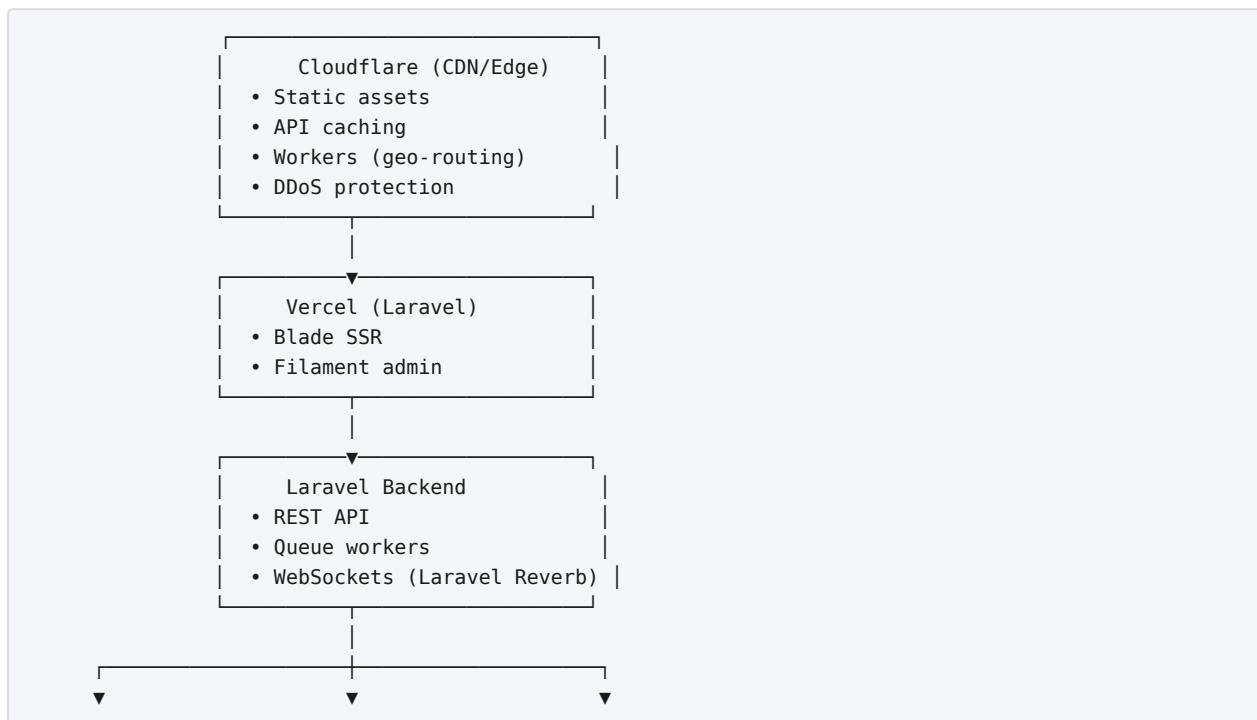
```

DB_CONNECTION=mysql DB_HOST=gateway01.us-west-2.prod.aws.tidbcloud.com
DB_PORT=4000 DB_DATABASE=kicc DB_USERNAME=xxxxx.root DB_PASSWORD=xxxxx

```

Zero code changes. Same Laravel migrations. Same Eloquent.

Integration Architecture



External APIs | R2 (Assets) | M-Pesa | 200+ tbls | Videos | Stripe |
 Auto-shard | Images | Africa Talking | HTAP | 3D HTML | Airlines GDS |
 Multi-reg | Backups | Shipping APIs |
 Google APIs | TiDB (DB) | Cloudflare |

Data Flow Examples

Tourist Books Kenyan Experience

1. User browses counties/tourism (Blade SSR via Vercel)
2. Agentic loop observes -> triggers recommendation refresh
3. AI pipeline scores user against: flights, hotels, restaurants, products
4. User sees personalized recommendations
5. Books: flight (GDS API) + hotel (TiDB) + restaurant (TiDB table booking)
6. Pipeline auto-books airport transfer (matching driver)
7. Payment: M-Pesa/Stripe -> TiDB transaction log
8. Order: Kenyan product recommended -> added to cart -> shipped
9. Agentic loop learns from booking -> updates match weights
10. Notification: SMS (Africa Talking) + email

Seller Lists Product

1. Seller from e.g., Kiambu lists avocado oil
2. Product written to TiDB (auto-sharded by product_id)
3. n8n workflow triggers: SEO metadata generation (agentic loop)
4. AI pipeline generates embedding vector -> stored in TiDB

5. Product visible in marketplace with global shipping calculation
6. TiFlash columnar replica enables real-time seller analytics
7. When ordered: inventory decremented, shipment created
8. Courier partner API called for pickup
9. Tracking events flow back through webhooks

Deployment Strategy

Phase 1 (Now) Phase 2 (Next) Phase 3 (Scale) ————— —————

————— • SQLite local • TiDB Serverless • TiDB Dedicated cluster • Laravel serve • Vercel deployment • Multi-region deployment • Basic features • E-commerce launch • Global CDN + edge • 56 tables • Travel booking • 200+ tables • 3D experiences • AI pipeline v1 • Full ML pipeline • n8n automation • Petabyte-scale • M-Pesa integration • 24/7 SRE

Project Structure

/home/kicc/Desktop/kicc/ |— kicc-platform/ ← NEW: Full self-hosted Laravel project | |— app/ PHP backend (32 models + 20 controllers + services) | |— database/ | |— schema.sql 56 current tables | |— schema-commerce.sql E-commerce + travel + ads + pipeline tables | |— schema-full.sql Complete ~200 table schema | |— public/3d/ Three.js experiences | |— setup.sh One-command setup | |— kenya-3d-platform/ ← EXISTING: Vercel deployment (unchanged) | |— laravel-backend/ Original Laravel for Vercel | |— pipeline/ Python pipeline (video gen, image analysis) | |— agentic_loop/ AI agent system | |— n8n-workflows/ Automation workflows | |— seo-engine/ SEO optimization | |— cloudflare-worker/ Edge functions | |— scripts/ Utility scripts | |— start-preview.sh Static preview server (port 8081)